

AI Hospital Appointment Prioritization

Project: AI Hospital Appointment Prioritization

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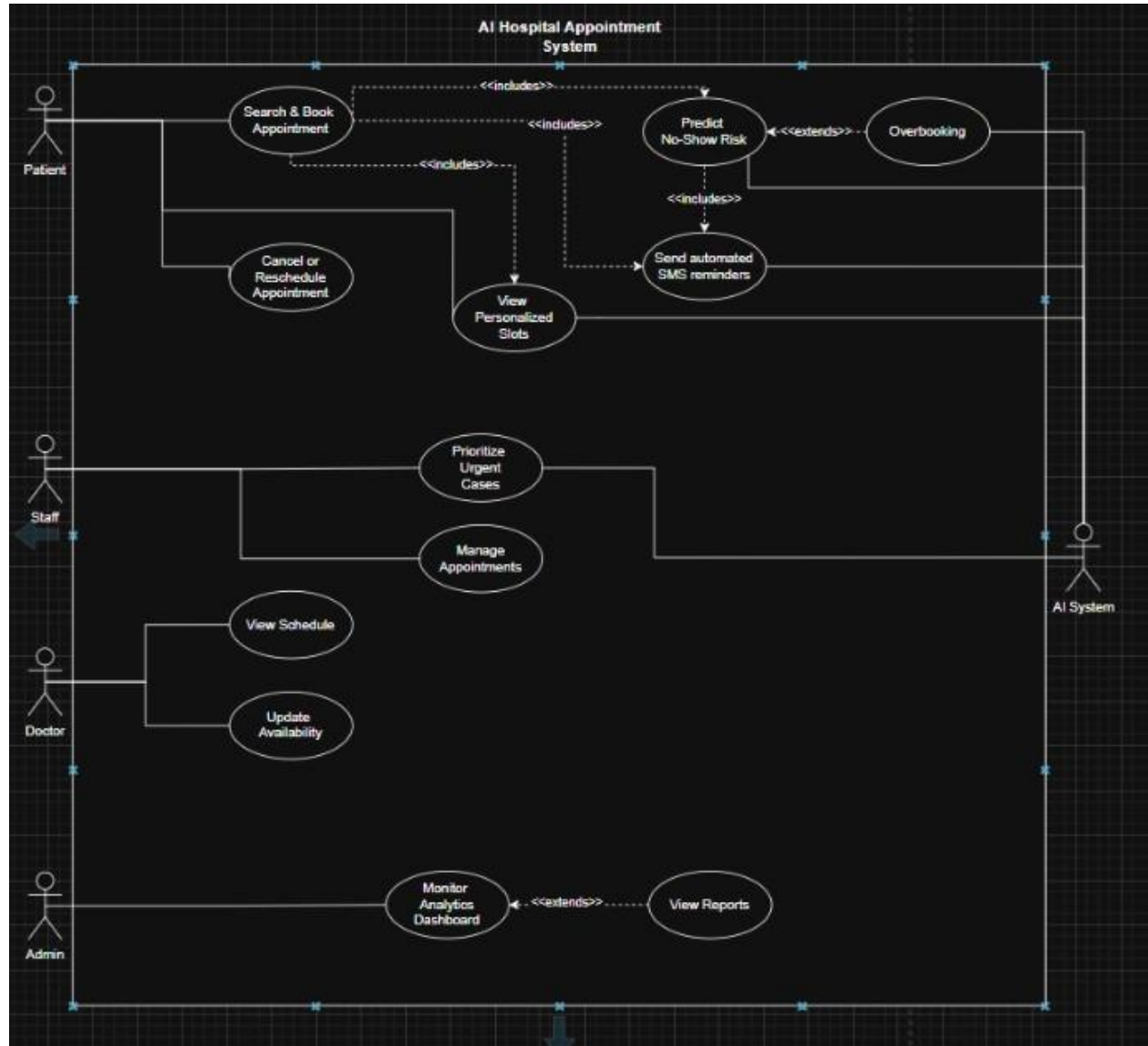
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Use Case Model

Use Case Diagram



Use Case Descriptions

Use Case Name: Search & Book Appointment

Actor: Patient

Description: Allows the patient to search for available appointment slots and book one based on AI recommendations.

Preconditions:

- Patient is registered and logged in

- System is operational

Main Flow:

1. Patient enters preferred date/time or doctor
2. System retrieves available slots
3. System shows **personalized slots** using AI
4. Patient selects a suitable slot
5. System checks availability
6. System confirms booking
7. System schedules SMS reminder

Alternative Flows:

- 2a. No slots available → system suggests alternative dates
- 4a. Selected slot becomes unavailable → system asks to choose another
- 6a. High no-show risk → system requests confirmation

Postconditions:

- Appointment is successfully booked
- Patient receives confirmation
- Reminder is scheduled

Use Case Name: Predict No-Show Risk

Actor: AI System

Description: The system predicts the probability of a patient missing their appointment using historical data.

Preconditions:

- Patient has appointment history
- AI model is trained and available

Main Flow:

1. System collects patient historical data
2. AI model analyzes attendance patterns
3. System calculates no-show probability
4. System labels risk level (low/medium/high)
5. System stores prediction result

Alternative Flows:

- 1a. No historical data → system assigns default risk
- 3a. Prediction fails → system logs error and continues without prediction

Postconditions:

- No-show probability is stored
- Result is used for reminders and scheduling decisions

Use Case Name: Cancel or Reschedule Appointment

Actor(s): Patient

Description: Allows the patient to cancel or reschedule an existing appointment.

Preconditions:

- Patient is logged in
- Patient has an existing appointment **Main Flow:**

1. Patient views upcoming appointments
2. Patient selects an appointment
3. Patient chooses cancel or reschedule
4. If reschedule → system shows available slots
5. Patient selects new slot (if rescheduling)
6. System updates appointment
7. System sends confirmation SMS

Alternative Flows:

- 3a. Cancellation too late → system prevents cancellation
- 4a. No slots available → system suggests alternatives
- 5a. Selected slot unavailable → patient chooses another

Postconditions:

- Appointment is canceled or updated
- System records the change
- Patient receives confirmation

Use Case Name: Manage Appointments

Actor: Staff (Receptionist)

Description: Allows staff to create, update, or manage patient appointments efficiently.

Preconditions:

- Staff is logged into the system
- System is connected to appointment database

Main Flow:

1. Staff accesses appointment dashboard
2. Staff searches for patient or schedule
3. Staff creates or edits appointment
4. System checks availability
5. System applies AI prioritization (urgent cases)
6. Staff confirms action
7. System saves changes

Alternative Flows:

- 3a. Invalid patient data → system shows error
- 4a. No available slots → system suggests alternatives
- 5a. Urgent case detected → system prioritizes earlier slot

Postconditions:

- Appointment is created/updated
- System reflects changes immediately
- Notifications may be triggered

Wireframes

- **P1 - Registration screen wireframe:**
- **Description:**
- Username field.
- Password field.
- email address field.
- Create account button.
- Login button.
- **Layout:**

	Username::	[	]	
	Password:	[	]	
	Email :	[	]	
	[	Create a new account	]	

	[	Login / you already have account	]	
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- **P2 - Login screen wireframe:**

- **Description:**

- Username field.
- Password field.
- Login button.
- Forgot password button.

- **Layout:**

	Patient Login	
	Username: []	
	Password: []	
	[Login]	

	[Forgot password]	
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- **P3 - View appointments:**

- **Description:**

- Select doctor field.

- Select date field.
- View available slots.
- **Layout:**

View Appointments
Doctor: [Drop List ▼]
Date: [Select Date]

Available Slots:
- 8:00 AM
- 10:00 AM

- **P4 - Book Appointment:**
- **Description:**
- Selected doctor.
- Selected time.
- Confirm booking.
- **Layout:**

Book Appointment
Doctor: [Dr.Ali]
Time slot: [8:00 AM]

[Confirm Booking]

- **P5 - Modify Appointment:**

- **Description:**

- Current doctor field.
- New doctor field.
- Current time field.
- New time field.
- Save changes button.

- **Layout:**

Modify Appointment
Current doctor: [Dr.Ali]
New doctor: [Drop List ▼]
Current time: [8:00 AM]
New time: [Select ▼]

[Save Changes]

- **P6 - Cancel Appointment:**

- **Description:**

- Appointment selection field.
- Are you sure Question.
- Yes and No buttons.

- **Layout:**

Cancel Appointment
Appointment: [8:00 AM]
Are You Sure ?
[Yes] [No]

- **P7 - Submit Patient Info:**

- **Description:**

- Name field.
- Age field.
- Symptoms field.
- Pain level field.
- Condition field.
- Vitals field.
- Submit button.

- **Layout:**

Patient Information	
Name:	[]
Age:	[]
Symptoms:	[]
Pain level:	[1-10]
Condition:	[Drop List ▼]
Vitals:	[]

[Submit]	

- **S1 - AI Patient Analysis:**

- **Description:**

- Input data processing.
- Model field.
- View result button.

- **Layout:**

	AI Analysis	
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	Input data processed [✓]	
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	Model: Prioritization AI	
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	[View Result]	
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- **S2 - Urgency Score:**

- **Description:**

- Patient field.
- Score field.
- Level field.

- **Layout:**

	Urgency Score	
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	Patient: [Ahmad]	
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	Score: [89%]	
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	Level: [High]	
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- **S3 - Appointment Prioritization:**

- **Description:**

- Names column.
- Condition column.
- Priority column.
- Time column.

- **Layout:**

	Priority Queue (AI)	
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Name	Condition	Priority	Time
-----	-----	-----	-----
Ahmad	Head Pain	Med	7 min
Ali	Ear Pain	High	20 min

- **S4 - Time Slot Recommendation:**

- **Description:**

- Based on urgency and availability.

- • Accept button.

- **Layout:**

Suggested Appointment Time
Based on urgency and availability

Today at 8:30 AM

[Accept]

- **S5 - No-Show Prediction:**

- **Description:**

- Patient name.

- Risk percent.

- Suggest reminder.

- **Layout:**

No-Show Prediction

Patient:	[Ali]
Risk:	[80%]

Suggest reminder	✓

- **S6 - Appointment Reminder:**
- **Description:**
- Appointment time.
- Notification status.\
- **Layout:**

Reminder Sent
Appointment at 12 PM
Notification sent ✓

- **S7 - Waitlist:**
- **Description:**
- Name column.
- Status column.
- **Layout:**

Waitlist
Name Status
----- -----
Sara Waiting
Omar Waiting

- **D1 - Doctor Dashboard:**
- **Description:**
- Priority Patients.
- Names of the patients.
- Priority level.
- View details button.
- **Layout:**

Doctor Dashboard	
Priority Patients:	
- Ali (High)	
- Ahmed (Med)	

[View Details]	

- **D2 - Patient History:**
- **Description:**
- Name of patient.
- Past visits for the patient.
- **Layout:**

Patient History	
Name: Ali	
Past Visits:	
- 13 Jan	
- 17 Feb	

- **R1 - Receptionist Dashboard:**

- **Description:**

- Todays appointments.
- Priority Queue.
- Manage button.

- **Layout:**

	Receptionist Dashboard	
	Todays appointments	
	Priority Queue	

	[Manage]	

- **R2 - Manage Appointments:**

- **Description:**

- Patient column.
- Time column.
- Action column.

- **Layout:**

Manage Appointments		
Patient	Time	Action
-----	-----	-----
Ali	8:30 AM	Edit / Delete

- **R3 - Register Patient:**

- **Description:**

- Name field.
- Age field.
- Save button.

- **Layout:**

Register Patient	
Name: []	
Age: []	-----
[Save]	

- **A1 - Admin Dashboard:**

- **Description:**

- No-show rate.
- Average waiting time.
- System usage.

- **Layout:**

Admin Dashboard	
No-show rate: 20%	
Average Waiting time: 15 min	
System Usage: High	

1. Extent of Alignment with User Needs

The identified requirements align with user needs to a high extent because each requirement was derived from a specific stakeholder bottleneck:

Patient Needs (Reduced Wait Times & Easy Access): The functional requirement for AI-Powered Slot Recommendations directly aligns with the need for faster access. By using data to prioritize, the system ensures that those with the highest medical need are not trapped in a first-come-first-served queue.

The Web Portal requirement addresses the need for 24/7 accessibility without calling the hospital.

Receptionist Needs (Efficiency & Reduced Workload): The Automated SMS Reminders and No-Show Prediction requirements align with the need to stop manual phone-call confirmations. This automates the most time-consuming part of their job.

Doctor Needs (Optimized Schedules): The non-functional requirement for 80% AI Accuracy ensures the system is reliable enough for doctors to trust the "overbooking" suggestions, meaning they have fewer empty slots and can see more patients per day.

Administrator Needs (Strategic Alignment): The Analytics Dashboard requirement aligns with the need to track KPIs that support the Qatar NHS-3, such as hospital resilience and service delivery excellence.

2. Effectiveness of the Designed IS (Information System)

The designed system meets these needs effectively by creating a "Feedback Loop" between data and action:

Predictive vs. Reactive: Most current systems are reactive (waiting for a patient to miss an appointment). This IS is effective because it is predictive; it identifies a high-risk noshow before it happens, allowing the system to offer that slot to a waitlisted patient.

Urgency-Based Logic: The effectiveness of the "Urgent Case Prioritization" feature is found in its ability to process complex patient symptoms (via the Submit Patient Info wireframe) into a simple Urgency Score. This bridges the gap between patient input and medical scheduling.

Non-Functional Resilience: By setting a requirement for System Scalability, the IS effectively prepares for Qatar's future population growth, ensuring the system doesn't slow down as more hospitals or patients are added.

Usability Targets: The design's effectiveness is validated by the 85% Satisfaction Target. The simple, low-complexity wireframes (P1-P7) ensure that even non-technical users can navigate the system easily, reducing the "Digital Divide."

